

5 Empirical Risk Minimization for Nonlinear Predictors

5.1 The Modern Regime

The results of the previous chapter are classical, and while a substantial body of work has since extended those results to the non-asymptotic, finite-sample regime, that work still makes the same structural assumptions as the classical theory. Namely, this literature assumes that

$$\hat{\theta}_T \longrightarrow \theta^* \tag{5.1}$$

is actually something that we care about! In particular, this presupposes that:

1. $h^*(x) = h(x; \theta^*)$ for some θ^* — i.e., the regression function is *known* up to a finite-dimensional parameter (Assumption 4.1 of the previous chapter);
2. $h(\cdot; \theta) = h(\cdot; \theta^*) \implies \theta = \theta^*$ — i.e., θ^* is *identifiable*;
3. θ^* corresponds to something we actually care about.

Before pushing on any of these, it is worth pausing to endorse what parameter recovery buys you, precisely because we are about to give it up. These guarantees state that $\hat{\theta}_T \approx \theta^*$ in the limit of large T . This implies **immediate generalization to new data**, as your estimate says something about the underlying physical (or behavioral) mechanism itself, not merely about how it behaves on the data you observed it on.

But modern approaches increasingly try to approximate h^* with an **overparameterized neural network** $h(\cdot; \theta)$. In this case the parameters θ are the weights of the network, and need not be uniquely defined, nor carry any semantic meaning. In this setting, assumptions 2 and 3 fail. Furthermore, we are often interested in trying to identify mappings that are “black-box” in nature, for example, to imitate a human in imitation learning, or to predict financial market behaviors. In such cases, assumption 1 fails as well.

Key Question 5.1. If there is no meaningful “true parameter,” what should identification even mean?

5.1.1 From Parameter Recovery to Prediction Guarantees

Instead of seeking to recover a true parameter θ^* , what if we instead only ask that the learned model $\hat{h}$ **predicts well**? As we will see over the rest of this chapter, the guarantees we can make under this weaker goal are qualitatively very different from — and quantitatively much weaker than — the classical parameter-recovery guarantees of the previous chapter. In exchange, though, the methods we develop are dramatically more broadly applicable: they impose essentially no structural assumptions on h^* at all, and apply just as well to an overparameterized neural network as to a linear model.

5.2 Statistical Learning Theory

The framework that formalizes “predicts well” is **statistical learning theory**, and its central object of study is **generalization**: the relationship between how well a model performs on the (seen) examples it was fit to, and how well it performs on (unseen) examples drawn from the same source. See Shalev-Shwartz and Ben-David [SB14] and Hardt and Recht [HR22] for a comprehensive treatment of the material developed in this chapter.

5.2.1 Risk Minimization

Setup. Let $(x, y) \in \mathcal{X} \times \mathcal{Y}$ be a pair of random variables with $(x, y) \sim D$ for some (unknown) distribution D . A **model** (or **predictor**) is a function $h : \mathcal{X} \rightarrow \mathcal{Y}$, and a **loss** is a function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$.

Definition 5.1 (Risk). The **risk** of a model h with respect to a loss ℓ is

$$R(h) := \mathbb{E}_{(x,y) \sim D} [\ell(h(x), y)]. \quad (5.2)$$

The learning problem we would ultimately like to solve is

$$\underset{h}{\text{minimize}} \quad \mathbb{E}_{(x,y) \sim D} [\ell(h(x), y)]. \quad (5.3)$$

By the tower property, this is the same as

$$\underset{h}{\text{minimize}} \quad \mathbb{E}_x [\mathbb{E}_{y|x} [\ell(h(x), y) \mid x]], \quad (5.4)$$

and, writing $p(x, y) = p(y \mid x)p(x)$ and unpacking the outer expectation as an integral,

$$\underset{h}{\text{minimize}} \quad \int_{\mathcal{X}} \mathbb{E}_{y|x} [\ell(h(x), y) \mid x] p(x) dx. \quad (5.5)$$

Because $p(x) \geq 0$ everywhere, this integral is minimized by minimizing the integrand *pointwise, for every x separately*: the optimization over the whole function h decouples into an independent scalar optimization at each x ,

$$h^*(x) = \arg \min_z \mathbb{E}_{y|x} [\ell(z, y) \mid x]. \quad (5.6)$$

Solving (5.6) directly would require: knowing $p(y \mid x)$; being able to solve $\int_{\mathcal{Y}} \ell(z, y) p_{y|x}(y \mid x) dy$ in closed form; and optimizing over *all* functions h , with no restriction on their form. None of these is available to us in practice — we know neither D nor $p(y \mid x)$, and “all functions” is not a class anyone can optimize over — so (5.3) is not directly solvable. Everything that follows is about what can be done instead, given only samples from D .

5.2.2 Training Data and Empirical Risk Minimization

Training set. Given a **training set** $S = ((x_1, y_1), \dots, (x_n, y_n)) \sim D^n$ of n i.i.d. draws from D , we define the **empirical risk** of a model h with respect to a loss ℓ as follows.

Definition 5.2 (Empirical risk).

$$R_S(h) := \frac{1}{n} \sum_{i=1}^n \ell(h(x_i), y_i). \quad (5.7)$$

Since S is random, $R_S(h)$ is a random variable depending on the training data S (for a *fixed*, non-random h).

Definition 5.3 (Empirical risk minimization). Given a hypothesis class $\mathcal{H}$ of candidate models, **empirical risk minimization (ERM)** is the problem

$$h_S^* := \arg \min_{h \in \mathcal{H}} R_S(h). \quad (5.8)$$

A few features of this definition are worth making explicit.

- $\mathcal{H}$ is typically chosen to be a hypothesis class that is “easy” to optimize over in practice — linear models, a fixed neural-network architecture, and so on — as opposed to the unrestricted “all functions h ” of (5.6).
- h_S^* is a random variable that depends, often in a very complicated way, on the training data S .
- h_S^* may not be exactly attained by the optimization algorithm used in practice, and it may not even be unique. To avoid confusion, we write $\hat{h}_S$ for whatever model the ERM procedure actually outputs, reserving h_S^* for an exact (possibly non-unique) minimizer.
- $R_S(\hat{h}_S)$ is called the **training error**, **training loss**, or **training risk**.

In the ideal case, the training risk $R_S(\hat{h}_S)$ would already be a good proxy for the true risk $R(\hat{h}_S)$; usually, however, $R_S(\hat{h}_S) < R(\hat{h}_S)$, since $\hat{h}_S$ was *chosen* to fit S well.

Definition 5.4 (Generalization error). The **generalization error** of a model h is the gap between the two,

$$\Delta_{\text{gen}}(h) := R(h) - R_S(h). \quad (5.9)$$

Rearranging the definition of Δ_{gen} gives what we will refer to, following the notes this chapter is drawn from, as the **fundamental theorem of ML**:

$$R(h) = \underbrace{R_S(h)}_{\text{we see this during training}} + \underbrace{\Delta_{\text{gen}}(h)}_{\text{we want to keep this small}}. \quad (5.10)$$

💡 *Key Result 5.1*

This is a tautology, not a theorem in any deep sense, but it correctly identifies the two pieces available to us. $R_S(h)$ is under our direct control: it is exactly what we minimize when we run ERM. $\Delta_{\text{gen}}(h)$ is not directly observable (it depends on the unknown D), and controlling it — for the *specific*, data-dependent model $\hat{h}_S$ that ERM outputs, not for some fixed h chosen in advance — is the entire technical content of the rest of this chapter.

5.2.3 A Detour on Bias/Variance and Overparameterization

Before developing tools to control Δ_{gen} , it is worth building the right mental model for how it interacts with the choice of hypothesis class $\mathcal{H}$. Since $\hat{h}_S$ approximately minimizes R_S over $\mathcal{H}$,

$$R_S(\hat{h}_S) \approx \min_{h \in \mathcal{H}} R_S(h), \quad (5.11)$$

suggesting the following heuristic dictionary: the achieved empirical risk $R_S(\hat{h}_S)$ behaves like the hypothesis class’s **bias** (how well the *best* element of $\mathcal{H}$ could possibly do), while some notion of the “complexity” of $\mathcal{H}$ behaves like its **variance** (its ability to overfit — to fit noise in S that will not be present in fresh data).

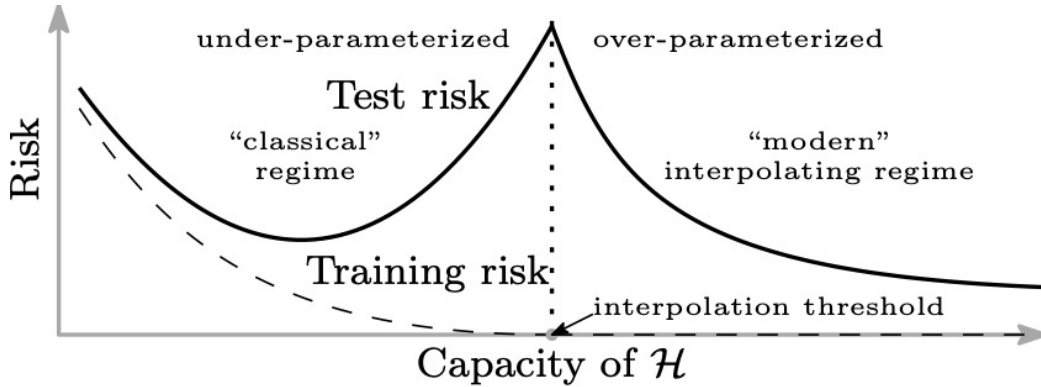


Figure 5.1: Risk vs. complexity($\mathcal{H}$): the classical bias/variance U-shape (underfit $\rightarrow$ sweet spot $\rightarrow$ overfit), followed — once complexity($\mathcal{H}$) crosses the **interpolation threshold** (roughly, complexity($\mathcal{H}$) = n , the point at which $\mathcal{H}$ is rich enough to fit the training data exactly) — by a **second descent**. Double descent is not universal: it does not appear for every hypothesis class and every data distribution, but it is also not a corner case restricted to a handful of contrived examples — it is the typical picture for sufficiently rich, sufficiently over-parameterized model classes to the right of the interpolation threshold.

The key empirical observation motivating this chapter, illustrated in fig. 5.1, is that in this “modern regime” the classical intuition

$$\Delta_{\text{gen}}(h) \propto \frac{\text{\#free parameters}}{\text{\#data points}} \quad (5.12)$$

is simply **vacuous**: heavily overparameterized models, with far more parameters than data points, routinely generalize *well* in practice, well past the interpolation threshold where the classical picture would predict catastrophic overfitting. See Belkin et al. [Bel+19] for the empirical case for double descent across a range of model classes.

Theories of generalization

Several theories of generalization have been developed. This chapter develops **uniform convergence** in depth; the others are mentioned here as pointers, and revisited briefly in the Further Reading.

- **Algorithmic stability.** Models output by a learning algorithm, $\hat{h}_S = \mathcal{A}(S)$, that are *insensitive* to small perturbations of the training data — e.g., $\|\mathcal{A}(S) - \mathcal{A}(S')\|$ is small whenever $\|S - S'\|$ is small — generalize well, essentially by definition: a model that cannot depend too strongly on any single training point cannot have memorized noise specific to that point. See Bousquet and Elisseeff [BE02].
- **Uniform convergence.** Generalization happens when $\mathcal{H}$ is “simple,” as measured by a complexity notion such as VC-dimension or Rademacher complexity. This is the route we develop below.
- **Margin bounds.** When the data is easily separable with a large margin, good generalization occurs.
- **Optimization.** Algorithmic choices in *how* ERM is solved (e.g., early stopping, the implicit bias of stochastic gradient descent) can by themselves yield models with good generalization properties, even within a hypothesis class that is nominally very rich.

Two scaling regimes

Anticipating the results to come, two qualitatively different rates appear depending on how well $\mathcal{H}$ fits the data:

1. When $R_S(h)$ is large (bounded away from 0), we expect $\Delta_{\text{gen}}(h) \lesssim 1/\sqrt{n}$.
2. When $R_S(h) \approx 0$ (the **realizable** or **interpolating** case), we expect the much faster $\Delta_{\text{gen}}(h) \lesssim 1/n$ — the “fast rate.”

We now derive both of these, in the simplest possible setting, before building up the general machinery.

5.3 Uniform Convergence

5.3.1 Warm-Up: A Single Binary Classifier

To see where the two rates above come from, study the behavior of a *single*, fixed $h \in \mathcal{H}$ under the 0–1 loss for binary classification,

$$\ell(h(x), y) = \mathbb{1}\{\text{sgn } h(x) \neq y\}, \quad y \in \{\pm 1\}. \quad (5.13)$$

Remark 5.5. Since $\ell(h(x), y) = \mathbb{1}\{\text{sgn } h(x) \neq y\}$ is a $\{0, 1\}$ -valued random variable,

$$\ell(h(x), y) \sim \text{Bernoulli}(p_h), \quad p_h := \mathbb{P}_D\{\text{sgn } h(x) \neq y\} \in [0, 1]. \quad (5.14)$$

Consequently

$$R(h) = \mathbb{E}_D[\mathbb{1}\{\text{sgn } h(x) \neq y\}] = p_h, \quad R_S(h) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\text{sgn } h(x_i) \neq y_i\}, \quad (5.15)$$

and, by the strong law of large numbers, $R_S(h) \xrightarrow{\text{a.s.}} R(h)$, i.e., $R_S(h) \rightarrow R(h)$ with probability 1.

The SLLN only tells us this happens *eventually*; we want a *quantitative* measure of how quickly.

Lemma 5.6 (Hoeffding’s inequality, [Hoe63]). *Suppose $\tilde{X}_1, \dots, \tilde{X}_n$ are independent random variables, with $\tilde{X}_i$ lying almost surely in $[a_i, b_i]$. Let $Z_n := \frac{1}{n} \sum_i \tilde{X}_i$. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}[Z_n \geq \mathbb{E}Z_n + \varepsilon] \leq \exp\left(\frac{-2n^2\varepsilon^2}{\sum_i (a_i - b_i)^2}\right). \quad (5.16)$$

Apply Lemma 5.6 with $\tilde{X}_i = -\mathbb{1}\{\text{sgn } h(x_i) \neq y_i\} \in [-1, 0]$, so that $Z_n = -R_S(h)$ and $\mathbb{E}Z_n = -R(h)$, and set $\varepsilon = \varepsilon$:

$$\mathbb{P}_{S \sim D^n}[R(h) - R_S(h) \geq \varepsilon] \leq \exp(-2n\varepsilon^2). \quad (5.17)$$

Inverting this bound by setting the right-hand side equal to δ , $\delta = \exp(-2n\varepsilon^2) \implies \varepsilon = \sqrt{\log(1/\delta)/2n}$, we conclude:

With probability at least $1 - \delta$ (over $S \sim D^n$), for a fixed h ,

$$\Delta_{\text{gen}}(h) = R(h) - R_S(h) \leq \sqrt{\frac{\log(1/\delta)}{2n}} \approx \frac{1}{\sqrt{n}}. \quad (5.18)$$

💡 Key Result 5.2

The realizable case. What if $R_S(h) = 0$? This happens exactly when $\text{sgn } h(x_i) = y_i$ for every $(x_i, y_i) \in S$. Fix $\varepsilon > 0$ and condition on $R(h) > \varepsilon$ (i.e., h is in fact a *bad* classifier). Then

$$\begin{aligned} \mathbb{P}_{S \sim D^n}[R_S(h) = 0 \mid R(h) > \varepsilon] &= \prod_{i=1}^n \mathbb{P}_{(x_i, y_i) \sim D}[\text{sgn } h(x_i) = y_i \mid R(h) > \varepsilon] \\ &= \prod_{i=1}^n \left(1 - \underbrace{\mathbb{P}_{(x_i, y_i) \sim D}[\text{sgn } h(x_i) \neq y_i \mid R(h) > \varepsilon]}_{> \varepsilon}\right) \\ &\leq \prod_{i=1}^n (1 - \varepsilon) \\ &= (1 - \varepsilon)^n \\ &\leq e^{-\varepsilon n}. \end{aligned}$$

That is, $\mathbb{P}\{R_S(h) = 0 \mid R(h) > \varepsilon\} \leq e^{-\varepsilon n}$ — in words, “the probability that S tricks us into thinking a bad classifier ($R(h) > \varepsilon$) is a good one ($R_S(h) = 0$)” decays *exponentially* in n . Inverting as before,

With probability at least $1 - \delta$, for a fixed h with $R_S(h) = 0$,

$$\Delta_{\text{gen}}(h) = R(h) - R_S(h) \leq \frac{\log(1/\delta)}{n} \approx \frac{1}{n}. \quad (5.19)$$

💡 Key Result 5.3

This is exactly the “fast rate” anticipated above: an extra factor of $1/\sqrt{n}$ is bought for free, purely by conditioning on the realizable case $R_S(h) = 0$.

Remark 5.7. Both bounds above are for a *fixed*, non-random h , chosen before seeing S . The challenge in carrying this analysis over to $\hat{h}_S$ — the model ERM actually outputs — is that $\hat{h}_S$ is *not* independent of S : it was chosen using S , so Hoeffding’s inequality does not apply to it directly.

5.3.2 Uniform Convergence over a Finite Hypothesis Class

The fix is to make the bound above hold *simultaneously* for every $h \in \mathcal{H}$, so that it applies in particular to whichever h the data happens to pick out.

Theorem 5.8 (Uniform convergence, finite $\mathcal{H}$). *Suppose $\mathcal{H}$ is finite, with $|\mathcal{H}|$ elements. Then with probability at least $1 - \delta$ (over $S \sim D^n$),*

$$\Delta_{\text{gen}}(h) \leq \sqrt{\frac{\log(|\mathcal{H}|/\delta)}{2n}} \quad \forall h \in \mathcal{H}. \quad (5.20)$$

Proof. For $\varepsilon > 0$, define the “bad event” for a given h , $\mathcal{E}_h := \{\Delta_{\text{gen}}(h) \geq \varepsilon\}$. Then, by a union bound,

$$\begin{aligned} \mathbb{P}_{S \sim D^n} [\exists h \in \mathcal{H} : \Delta_{\text{gen}}(h) > \varepsilon] &= \mathbb{P}_{S \sim D^n} \left[\bigcup_{h \in \mathcal{H}} \mathcal{E}_h \right] \\ &\leq \sum_{h \in \mathcal{H}} \mathbb{P}_{S \sim D^n} (\mathcal{E}_h) \\ &= \sum_{h \in \mathcal{H}} \mathbb{P}_{S \sim D^n} [\Delta_{\text{gen}}(h) > \varepsilon] \\ &\leq |\mathcal{H}| \exp(-2n\varepsilon^2), \end{aligned}$$

using Key Result 5.2 on each individual term. Inverting (setting the right-hand side to δ) gives the claim. $\square$

In general, bounds of this flavor will look like $\sqrt{\text{complexity}(\mathcal{H})/n}$, with $\log |\mathcal{H}|$ playing the role of the “complexity” term for a finite class. To generalize this to classes $\mathcal{H}$ of infinite cardinality — linear predictors, neural networks, and so on — we need a notion of complexity that remains finite even when $|\mathcal{H}| = \infty$. Several such measures exist: VC-dimension, covering numbers, and Rademacher (or Gaussian) complexity. We develop Rademacher complexity next, and covering numbers in section 5.5.

5.4 Rademacher Complexity

Write $z_i := (x_i, y_i)$, so that $S = (z_i)_{i=1}^n \sim D^n$. Let $\{\sigma_i\}_{i=1}^n$ be i.i.d. **Rademacher random variables**, i.e., $\sigma_i = +1$ with probability $\frac{1}{2}$ and $\sigma_i = -1$ with probability $\frac{1}{2}$, independent of S .

We state the definitions and the main uniform-convergence theorem of this section for a *generic*, abstract class $\mathcal{G}$ of bounded real-valued functions of $z = (x, y)$ — not yet the loss-composed-with- predictor class we actually care about — to keep the statement fully general; we specialize it to our setting immediately afterward.

Definition 5.9 (Rademacher complexity). For a class $\mathcal{G}$ of functions $g : \mathcal{Z} \rightarrow \mathbb{R}$, the **empirical Rademacher complexity** of $\mathcal{G}$ (with respect to S) is

$$\hat{R}_n(\mathcal{G}) := \mathbb{E}_\sigma \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^n \sigma_i g(z_i) \right], \quad (5.21)$$

and the **Rademacher complexity** of $\mathcal{G}$ is its average over draws of S ,

$$R_n(\mathcal{G}) := \mathbb{E}_{z_i \sim D} [\hat{R}_n(\mathcal{G})]. \quad (5.22)$$

Intuitively, $R_n(\mathcal{G})$ measures how well functions in $\mathcal{G}$ can, on average, be made to correlate with pure random noise ($\{\sigma_i\}$) evaluated at n sample points — a rich class can correlate well with almost any sign pattern, a simple one cannot.

Theorem 5.10 (Uniform convergence via Rademacher complexity). *Fix a failure probability $\delta \in (0, 1)$, and suppose $\mathcal{G} \subseteq \{g : \mathcal{Z} \rightarrow [0, B]\}$ and $S = \{z_1, \dots, z_n\} \sim D^n$. Then, with probability at least $1 - \delta$ over $S \sim D^n$, simultaneously for every $g \in \mathcal{G}$,*

$$\mathbb{E}_{z \sim D} g(z) \leq \frac{1}{n} \sum_{i=1}^n g(z_i) + 2R_n(\mathcal{G}) + B \sqrt{\frac{\log(1/\delta)}{2n}}. \quad (5.23)$$

Proof. Step 1: reduce to bounding a single supremum. For any $g \in \mathcal{G}$,

$$\begin{aligned} \mathbb{E}_{z \sim D} g(z) &= \frac{1}{n} \sum_{i=1}^n g(z_i) + \left(\mathbb{E}_{z \sim D} g(z) - \frac{1}{n} \sum_{i=1}^n g(z_i) \right) \\ &\leq \frac{1}{n} \sum_{i=1}^n g(z_i) + \sup_{g \in \mathcal{G}} \left[\mathbb{E}_{z \sim D} g(z) - \frac{1}{n} \sum_{i=1}^n g(z_i) \right]. \end{aligned}$$

Define the random variable

$$\Psi(S) := \sup_{g \in \mathcal{G}} \left[\mathbb{E}_{z \sim D} g(z) - \frac{1}{n} \sum_{i=1}^n g(z_i) \right]. \quad (5.24)$$

It remains to show that $\Psi(S)$ concentrates around its mean $\mathbb{E}\Psi(S)$.

Step 2: McDiarmid's inequality.

Lemma 5.11 (Bounded differences, [McD89]). *Let $z_1, \dots, z_n$ be independent random variables taking values in $\mathcal{Z}$, and suppose $\Psi : \mathcal{Z}^n \rightarrow \mathbb{R}$ satisfies the bounded-differences property*

$$\sup_{(z_1, \dots, z_n), \hat{z}_i} |\Psi(z_1, \dots, z_n) - \Psi(z_1, \dots, z_{i-1}, \hat{z}_i, z_{i+1}, \dots, z_n)| \leq c_i \quad \forall i = 1, \dots, n. \quad (\star)$$

Then, for every $t > 0$,

$$\mathbb{P}_{S \sim D^n} [\Psi(z_1, \dots, z_n) \geq \mathbb{E}\Psi + t] \leq \exp\left(\frac{-2t^2}{\sum_i c_i^2}\right). \quad (5.25)$$

(Hoeffding's inequality, Lemma 5.6, is itself a special case of McDiarmid's inequality, applied to $\Psi(z_1, \dots, z_n) = \frac{1}{n} \sum_i z_i$.) Since $\mathcal{G} \subseteq \{g : \mathcal{Z} \rightarrow [0, B]\}$, changing a single z_i changes $\Psi(S)$ by at most B/n , so $(\star)$ holds with $c_i \equiv B/n$. Applying Lemma 5.11 to Ψ , we conclude that with probability at least $1 - \delta$,

$$\mathbb{E}_{z \sim D} g(z) \leq \underbrace{\frac{1}{n} \sum_{i=1}^n g(z_i) + \mathbb{E}_{S \sim D^n} \left[\sup_{g \in \mathcal{G}} \mathbb{E}_{z \sim D} g(z) - \frac{1}{n} \sum_{i=1}^n g(z_i) \right]}_{\text{last thing to deal with: } \mathbb{E}\Psi(S)} + B \sqrt{\frac{\log(1/\delta)}{2n}}. \quad (5.26)$$

Step 3: symmetrization. It remains to bound $\mathbb{E}\Psi(S)$ itself by $2R_n(\mathcal{G})$. The trick is to introduce an independent **ghost sample** $\bar{S} = (\bar{z}_1, \dots, \bar{z}_n) \sim D^n$, drawn independently of S . Since $\mathbb{E}_{z \sim D} g(z) = \mathbb{E}_{\bar{S} \sim D^n} [\frac{1}{n} \sum_i g(\bar{z}_i)]$, while $\frac{1}{n} \sum_i g(z_i)$ does not depend on $\bar{S}$ at all (so it trivially also equals $\mathbb{E}_{\bar{S} \sim D^n} [\frac{1}{n} \sum_i g(z_i)]$),

$$\begin{aligned} \mathbb{E}\Psi(S) &= \mathbb{E}_{S \sim D^n} \left[\sup_{g \in \mathcal{G}} \mathbb{E}_{z \sim D} g(z) - \frac{1}{n} \sum_i g(z_i) \right] \\ &= \mathbb{E}_{S \sim D^n} \left[\sup_{g \in \mathcal{G}} \mathbb{E}_{\bar{S} \sim D^n} \frac{1}{n} \sum_i (g(\bar{z}_i) - g(z_i)) \right] \\ &\leq \mathbb{E}_{S, \bar{S} \sim D^n \times D^n} \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_i (g(\bar{z}_i) - g(z_i)) \right], \end{aligned} \quad (5.27)$$

where the inequality moves the supremum outside the (inner) expectation over $\bar{S}$ (Jensen's inequality for the convex function $\sup(\cdot)$). Now, since S and $\bar{S}$ are i.i.d. and independent of each other, swapping $z_i \leftrightarrow \bar{z}_i$ for any subset of indices leaves the joint distribution of (5.27) unchanged; equivalently, multiplying $(g(\bar{z}_i) - g(z_i))$ by an independent Rademacher sign σ_i does not change its distribution. Hence

$$\begin{aligned} \mathbb{E}_{S, \bar{S}} \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_i (g(\bar{z}_i) - g(z_i)) \right] &= \mathbb{E}_{S, \bar{S}, \sigma} \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_i \sigma_i (g(\bar{z}_i) - g(z_i)) \right] \\ &\leq \mathbb{E}_{\bar{S}, \sigma} \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_i \sigma_i g(\bar{z}_i) \right] + \mathbb{E}_{S, \sigma} \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_i (-\sigma_i) g(z_i) \right] \\ &= \underbrace{\mathbb{E}_{\bar{S}, \sigma} \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_i \sigma_i g(\bar{z}_i) \right]}_{= R_n(\mathcal{G})} + \underbrace{\mathbb{E}_{S, \sigma} \left[\sup_{g \in \mathcal{G}} \frac{1}{n} \sum_i (-\sigma_i) g(z_i) \right]}_{= R_n(\mathcal{G})} \\ &= 2R_n(\mathcal{G}), \end{aligned}$$

where the inequality splits the supremum of a sum into a sum of suprema (subadditivity), and the last line uses that $\{-\sigma_i\}$ has the same distribution as $\{\sigma_i\}$, so both terms equal $R_n(\mathcal{G})$ by definition. Combining with (5.26) gives (5.23). $\square$

5.4.1 Useful Properties of Rademacher Complexity

Two structural facts make Rademacher complexity easy to work with in practice.

Proposition 5.12 (Scaling). *For a function class $\mathcal{G}$ and constants $a, b \in \mathbb{R}$, let $a\mathcal{G} + b := \{z \mapsto a g(z) + b : g \in \mathcal{G}\}$. Then $R_n(a\mathcal{G} + b) = |a| R_n(\mathcal{G})$: additive shifts do not change the Rademacher complexity (since $\mathbb{E}_\sigma[\sum_i \sigma_i] = 0$), and scaling by a scales it by $|a|$.*

Proposition 5.13 (Contraction / Ledoux–Talagrand inequality, [LT91]). *Let $\mathcal{G}$ be a function class and let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be L_φ -Lipschitz with $\varphi(0) = 0$. Then $R_n(\varphi \circ \mathcal{G}) \leq 2L_\varphi R_n(\mathcal{G})$.*

This last fact is genuinely not obvious — composing with a Lipschitz function only *doubles* the Rademacher complexity, rather than scaling it by some quantity that could grow with n or the richness of $\mathcal{G}$ — and is what lets us pass Rademacher complexity bounds through a loss function, as the next example shows.

5.4.2 Example: Binary Classification with Hinge Loss

We now specialize Theorem 5.10 to a concrete setting, applying it to the class $\mathcal{G} = \ell \circ \mathcal{H} := \{\ell_h : h \in \mathcal{H}\}$, where $\ell_h(z) := \ell(h(x), y)$, so that $\frac{1}{n} \sum_i \ell_h(z_i) = R_S(h)$ and $\mathbb{E}_{z \sim D} \ell_h(z) = R(h)$, i.e., Theorem 5.10 directly bounds $\Delta_{\text{gen}}(h)$ for every $h \in \mathcal{H}$ at once.

Example 5.14 (Binary classification with hinge loss). Let $\ell_h(z) = \ell(h(x), y) = [1 - yh(x)]_+$ be the **hinge loss**, $\mathcal{Y} = \{\pm 1\}$, and define $B_\ell := \sup_{x \in \mathcal{X}, y \in \mathcal{Y}, h \in \mathcal{H}} \ell(h(x), y)$. By Theorem 5.10 applied to $\mathcal{G} = \ell \circ \mathcal{H}$, with probability at least $1 - \delta$, for every $h \in \mathcal{H}$,

$$\mathbb{E}_{z \sim D} \ell_h(z) \leq \frac{1}{n} \sum_{i=1}^n \ell_h(z_i) + 2R_n(\ell \circ \mathcal{H}) + B_\ell \sqrt{\frac{\log(1/\delta)}{2n}}. \quad (5.28)$$

Step 1: bound $R_n(\ell \circ \mathcal{H})$ via contraction. Note that $\ell \circ \mathcal{H} = [\tilde{\ell} \circ \mathcal{H}]_+$, where $\tilde{\ell}_h(z) := 1 - yh(x)$ (the “raw margin,” before the hinge). Since $\varphi(z) := [z]_+$ satisfies $\varphi(0) = 0$ and $\text{Lip}(\varphi) = 1$, Proposition 5.13 gives $R_n(\ell \circ \mathcal{H}) \leq 2R_n(\tilde{\ell} \circ \mathcal{H})$.

Now take $\mathcal{H} = \{h(x) = w^\top x : \|w\| \leq W\}$, the class of linear predictors of norm at most W , and assume $\|x\|_2 \leq B_{\mathcal{X}}$ almost surely.

Step 1a: bound B_ℓ .

$$\begin{aligned} B_\ell &= \sup_{\|x\|_2 \leq B_{\mathcal{X}}, y \in \{\pm 1\}, \|w\| \leq W} 1 - yw^\top x \\ &= 1 + \sup_{\|x\| \leq B_{\mathcal{X}}, \|w\| \leq W} w^\top x \\ &= 1 + B_{\mathcal{X}}W. \end{aligned}$$

Step 1b: bound $R_n(\mathcal{H})$.

$$\begin{aligned} R_n(\mathcal{H}) &= \frac{1}{n} \mathbb{E}_{S, \sigma} \left[\sup_{\|w\| \leq W} \sum_{i=1}^n \sigma_i w^\top x_i \right] \\ &= \frac{1}{n} \mathbb{E}_{S, \sigma} \left[\sup_{\|w\| \leq W} w^\top \left(\sum_i \sigma_i x_i \right) \right] \\ &= \frac{W}{n} \mathbb{E}_{S, \sigma} \left\| \sum_i \sigma_i x_i \right\|_2 \quad (\text{sup over a ball is achieved by Cauchy-Schwarz}) \\ &= \frac{W}{n} \mathbb{E}_{S, \sigma} \sqrt{\sum_{i,j} \sigma_i \sigma_j x_i^\top x_j} \\ &\stackrel{\text{Jensen}}{\leq} \frac{W}{n} \sqrt{\mathbb{E}_{S, \sigma} \sum_{i,j} \sigma_i \sigma_j x_i^\top x_j} \\ &= \frac{W}{n} \sqrt{\mathbb{E}_S \sum_i \|x_i\|_2^2} \quad (\mathbb{E}[\sigma_i \sigma_j] = 0 \text{ for } i \neq j, \sigma_i^2 = 1) \\ &\leq \frac{W}{n} \sqrt{\sum_{i=1}^n B_{\mathcal{X}}^2} \\ &= \frac{WB_{\mathcal{X}}}{\sqrt{n}}. \end{aligned}$$

Combining with the contraction step, $R_n(\ell \circ \mathcal{H}) \leq 2R_n(\tilde{\ell} \circ \mathcal{H}) = 2R_n(\mathcal{H}) \leq 2WB_{\mathcal{X}}/\sqrt{n}$ (translating by the constant 1 inside $\tilde{\ell}$ does not change R_n , by the scaling property).

Putting it together. With probability at least $1 - \delta$, for every $h \in \mathcal{H}$,

$$\Delta_{\text{gen}}(h) \leq \frac{4WB_{\mathcal{X}}}{\sqrt{n}} + (1 + B_{\mathcal{X}}W)\sqrt{\frac{\log(1/\delta)}{2n}} \lesssim \frac{B_{\mathcal{X}}W}{\sqrt{n}}, \quad (5.29)$$

i.e., the norm bound W — *not* the (infinite) cardinality of $\mathcal{H}$ — is the right measure of the complexity of this hypothesis class.

5.5 Covering Numbers and Dudley’s Entropy Integral

The hinge-loss example above required us to bound $R_n(\mathcal{H})$ by hand, using the specific linear structure of $\mathcal{H}$. We now develop a general tool — **covering numbers** and **Dudley’s entropy integral** — that bounds $R_n(\mathcal{H})$ for essentially any class $\mathcal{H}$ equipped with a metric, reducing the Rademacher-complexity question to a purely geometric one.

Definition 5.15 (Covering number). Let $(\mathcal{X}, \rho)$ be a metric space. A set $\{x_i\}_{i=1}^N \subset \mathcal{X}$ is an ε -**net** (or ε -**covering**) of $\mathcal{X}$ if for every $x \in \mathcal{X}$ there is some x_i with $\rho(x, x_i) \leq \varepsilon$. The **covering number** $N(\varepsilon; \mathcal{X}, \rho)$ is the cardinality of the smallest such ε -net.

Proposition 5.16 (VC-dimension controls covering numbers). *If $\text{VC-dim}(\mathcal{H}) = v$, then there exist universal constants $C, c > 0$ such that $N(\varepsilon; \mathcal{H}, \|\cdot\|_n) \leq Cv(c/\varepsilon)^v$, and hence $R_n(\mathcal{H}) \lesssim \sqrt{v/n}$. See Wainwright [Wai19, Ch. 5] (building on the original Vapnik and Chervonenkis [VC68]) for the exact constants and a proof.*

Theorem 5.17 (Dudley’s entropy integral, simplified; cf. [Wai19, Ch. 5]). *Let $\{X_\theta, \theta \in \Pi\}$ be a zero-mean, bounded stochastic process indexed by $\theta \in \Pi$, with pseudometric $\rho_X(\theta, \tilde{\theta}) := (\mathbb{E}[(X_\theta - X_{\tilde{\theta}})^2])^{1/2}$, and let $\mathfrak{D} := \sup_{\theta, \tilde{\theta} \in \Pi} \rho_X(\theta, \tilde{\theta})$ be the “diameter” of the process. Define*

$$J_{\mathfrak{D}} := \int_0^{\mathfrak{D}} \sqrt{\log N(u; \Pi, \rho_X)} \, du. \quad (5.30)$$

Then

$$\mathbb{E} \left[\sup_{\theta, \tilde{\theta} \in \Pi} (X_\theta - X_{\tilde{\theta}}) \right] \leq 16 J_{\mathfrak{D}}. \quad (*)$$

In particular, if $0 \in \Pi$ (so we may take $\tilde{\theta}$ fixed at $X_0 = 0$), $(*)$ gives $\mathbb{E}[\sup_{\theta} X_\theta] \leq 16 J_{\mathfrak{D}}$ directly.

5.5.1 Application to Rademacher Complexity

Fix S , define the zero-mean random variable $Z_h := \frac{1}{\sqrt{n}} \sum_i \sigma_i h(x_i)$ for $h \in \mathcal{H}$, and let $\{Z_h, h \in \mathcal{H}\}$ be our stochastic process (indexed by $\mathcal{H}$ itself, playing the role of Π above). Assume additionally that $|h(x)| \leq b$ for all $x \in \mathcal{X}$ and $h \in \mathcal{H}$, and define the empirical

L^2 -pseudometric

$$\rho_X^2(h, h') := \|h - h'\|_n^2 := \frac{1}{n} \sum_{i=1}^n \|h(x_i) - h'(x_i)\|^2 \leq 4b^2. \quad (5.31)$$

Applying (*) with diameter $\mathfrak{D} = 2b$,

$$\hat{R}_n(\mathcal{H}) = \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_i \sigma_i h(x_i) \right] = \frac{1}{\sqrt{n}} \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} Z_h \right] \leq \frac{16}{\sqrt{n}} \int_0^{2b} \sqrt{\log N(u; \mathcal{H}, \|\cdot\|_n)} du. \quad (5.32)$$

Bounding $R_n(\mathcal{H})$ has been reduced to bounding covering numbers $N(\varepsilon; \mathcal{H}, \|\cdot\|_n)$: (5.32) converts *any* bound on how many ε -balls it takes to cover $\mathcal{H}$ into a Rademacher complexity bound, with no further reference to the specific structure of $\mathcal{H}$.

💡 Key Result 5.4

5.5.2 Properties of Covering Numbers

Three elementary properties let us actually compute covering numbers for parametric classes.

Proposition 5.18 (Covering number of a Euclidean ball). $\log N(\varepsilon; B_2^n(1), \|\cdot\|_2) \leq n \log(1 + 2/\varepsilon)$, and hence $N(\varepsilon; B_2^n(1), \|\cdot\|_2) \lesssim (1/\varepsilon)^n$.

Proposition 5.19 (Covering numbers and Lipschitz reparameterization). *Let $h_\theta \in \mathcal{H}$ be indexed by $\theta \in \Theta$, let $\|\cdot\|_{\mathcal{H}}$ be a metric on $\mathcal{H}$ and $\|\cdot\|_{\Theta}$ a metric on Θ , and suppose $\theta \mapsto h_\theta$ is L_θ -Lipschitz, i.e., $\|h_{\theta_1} - h_{\theta_2}\|_{\mathcal{H}} \leq L_\theta \|\theta_1 - \theta_2\|_{\Theta}$. Then*

$$N(\varepsilon; \mathcal{H}, \|\cdot\|_{\mathcal{H}}) \leq N(\varepsilon/L_\theta; \Theta, \|\cdot\|_{\Theta}). \quad (5.33)$$

Proof. Let $\{\theta_i\}_{i=1}^{N_\Theta}$ be an (ε/L_θ) -net of Θ , and define $\{h_{\theta_i}\}_{i=1}^{N_\Theta}$. For any $h_\theta \in \mathcal{H}$, choose h_{θ^*} from this set with $\|\theta - \theta^*\|_{\Theta} \leq \varepsilon/L_\theta$. Then $\|h_\theta - h_{\theta^*}\|_{\mathcal{H}} \leq L_\theta \|\theta - \theta^*\|_{\Theta} \leq L_\theta \cdot \varepsilon/L_\theta = \varepsilon$, so $\{h_{\theta_i}\}_{i=1}^{N_\Theta}$ is an ε -net of $\mathcal{H}$, giving $N(\varepsilon; \mathcal{H}, \|\cdot\|_{\mathcal{H}}) \leq N_\Theta = N(\varepsilon/L_\theta; \Theta, \|\cdot\|_{\Theta})$. $\square$

Proposition 5.20 (Covering a bounded parameter set). *If $\Theta, B_2^k(1) \subset \mathbb{R}^k$ and $\|\theta\|_2 \leq B_\Theta$ for every $\theta \in \Theta$, then*

$$N(\varepsilon; \Theta, \|\cdot\|_2) \leq N(\varepsilon/B_\Theta; B_2^k(1), \|\cdot\|_2). \quad (5.34)$$

Proof. Let $\{x_i\}_{i=1}^{N_2}$ be an (ε/B_Θ) -net of $B_2^k(1)$, and define $\{z_i\}_{i=1}^{N_2} := \{B_\Theta x_i\}_{i=1}^{N_2}$; since $\|z_i\|_2 = B_\Theta \|x_i\|_2 \leq B_\Theta$, we have $\{z_i\} \subset \Theta$ (as long as Θ contains the full ball of radius B_Θ ; otherwise this still gives a valid, possibly non-tight, cover of $\Theta \subseteq B_\Theta B_2^k(1)$). For an arbitrary $\theta \in \Theta$, $\theta/B_\Theta \in B_2^k(1)$, so there is $x^* \in \{x_i\}$ with $\|\theta/B_\Theta - x^*\| \leq \varepsilon/B_\Theta$, and hence $\|\theta - B_\Theta x^*\| = B_\Theta \|\theta/B_\Theta - x^*\| \leq B_\Theta \cdot \varepsilon/B_\Theta = \varepsilon$. $\square$

5.6 Application: Learning Predictors for Dynamical Systems

We now bring the machinery of this chapter back to the setting of the previous chapter: learning a one-step predictor for a dynamical system, via ERM rather than parametric prediction-error estimation. The emphasis throughout is on **the effect of the dynamical system's stability on sample complexity** — exactly the theme first previewed in the stability sidebar of section 4.3.2.

Remark 5.21 (Connection to chapter 5's predecessor). This section develops the system identification case explicitly. As noted at the start of the previous chapter, the same general time-series model (4.1) covers imitation learning as well, with the covariates generated along expert trajectories rather than by the dynamics directly (chapter 4's two motivating examples). Every result below applies to that setting essentially unchanged — the only thing that differs is how the state sequence $\{x_t\}$ is generated in the first place.

Setup. Consider a dynamical system

$$x_{t+1} = h^*(x_t), \quad x_0 = \xi \sim D, \quad \|x_0\| \leq B_0 \text{ a.s.} \quad (5.35)$$

Our **goal** is to learn a predictor $\hat{h}$ such that $x_{t+1} \approx \hat{h}(x_t)$ for trajectories started from fresh $x_0 \sim D$.

Data. We observe n i.i.d. trajectories $\{\xi_i, x_1^{(i)}(\xi_i), \dots, x_T^{(i)}(\xi_i)\}_{i=1}^n$, generated by drawing n i.i.d. initial conditions $\{\xi_i\}_{i=1}^n \sim D^n$ and rolling (5.35) forward. We find $\hat{h}$ by solving the ERM problem

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \sum_{t=0}^{T-1} \ell(h(x_t^{(i)}), x_{t+1}^{(i)}), \quad \text{e.g., } \ell(h(x_t), x_{t+1}) = \|x_{t+1} - h(x_t)\|_2. \quad (5.36)$$

Note the shift in perspective from chapter 5's predecessor: each *trajectory* ξ_i (rather than each *time step*) is now one i.i.d. sample, and the loss being averaged over trajectories, $L(\xi; h) := \sum_{t=0}^{T-1} \ell(h(x_t), x_{t+1})$, is itself a sum over the T steps of a single rollout.

5.6.1 Bounding the Trajectory Loss

Applying Theorem 5.10 to (5.36) requires $L(\cdot; h)$ to be uniformly bounded over $\|\xi\| \leq B_0$ and $h \in \mathcal{H}$. We make the following regularity assumptions.

- Assumption 5.22** (Regularity for trajectory ERM). (i) $\xi \sim D$ and $\|\xi\| \leq B_0$ almost surely.
- (ii) The per-step loss is Lipschitz-controlled by its arguments: $|\ell(z, y)| \leq L_\ell(\|z\| + \|y\|)$.
- (iii) Every candidate predictor shares a common Lipschitz bound with the true dynamics: $\|h(x)\| \leq L_h \|x\|$ for all $h \in \mathcal{H}$, and $\mathcal{H} = \{h : \text{Lip}(h) \leq L_h\}$.

Assumption (iii) is a realizability-style condition: it says every candidate in $\mathcal{H}$, and (implicitly, since we want the bound to apply along the actual trajectory) the true map h^* itself, obeys the same Lipschitz bound L_h . Under it, $\|x_t\| \leq L_h^t \|x_0\| \leq L_h^t B_0$ almost surely, by induction on (5.35).

A first, assumption-free bound. Combining (i)–(iii),

$$\begin{aligned}
 \sup_{\|\xi\| \leq B_0, h \in \mathcal{H}} L(\xi; h) &\leq \sup_* L_\ell \sum_{t=0}^{T-1} (\|h(x_t)\| + \|x_{t+1}\|) \\
 &\leq \sup_* L_\ell \sum_{t=0}^{T-1} (L_h \|x_t\| + \|x_{t+1}\|) \\
 &\leq 2L_\ell \sum_{t=0}^{T-1} L_h^{t+1} B_0 \\
 &= 2L_\ell L_h B_0 \sum_{t=0}^{T-1} L_h^t \\
 &= 2L_\ell L_h B_0 \frac{L_h^T - 1}{L_h - 1} \\
 &= \tilde{O}(L_h^T).
 \end{aligned} \tag{5.37}$$

With no stability assumption at all, the loss bound (and, as we will see below, the generalization bound built on it) grows *exponentially* in the prediction horizon T .

Case 2: exponential stability. Suppose instead h^* is exponentially stable along the trajectory: there exist $\beta > 0$, $\rho \in [0, 1)$ such that $\|x_t\| \leq \beta \rho^t \|x_0\| \leq \beta B_0 \rho^t$. Then

$$\begin{aligned}
 \sum_{t=0}^{T-1} \|x_t\| &\leq \beta B_0 \sum_{t=0}^{T-1} \rho^t \\
 &\leq \frac{\beta B_0}{1 - \rho} \\
 &= \tilde{O}(1),
 \end{aligned}$$

i.e., with exponential stability the loss bound is **dimension- and horizon-independent**: $B_\ell = \tilde{O}(1)$.

Case 3: polynomial ($\mathcal{KL}$ -type) stability. More generally, suppose $\|x_t\| \leq \beta(\|x_0\|, t)$ for β a class $\mathcal{KL}$ function (definition 3.5) — this interpolates between the two cases above. Take, concretely, $\beta(s, t) = \tilde{\beta}s/(1+t)^\alpha$ for some rate $\alpha > 0$. Then

$$\begin{aligned}
 \sum_{t=0}^{T-1} \beta(\|x_0\|, t) &\leq \tilde{\beta} B_0 \sum_{t=0}^{T-1} \frac{1}{(1+t)^\alpha} \\
 &\leq \tilde{\beta} B_0 \left(1 + \int_0^{T-1} \frac{dt}{(1+t)^\alpha}\right),
 \end{aligned}$$

by comparing the sum to an integral. Evaluating the integral gives three regimes, depending on the polynomial rate α :

$$\sum_{t=0}^{T-1} \beta(\|x_0\|, t) \lesssim \begin{cases} \tilde{\beta} B_0 \frac{T^{1-\alpha}}{1-\alpha} = \tilde{O}(T^{1-\alpha}), & \alpha \in (0, 1), \\ \tilde{\beta} B_0 (1 + \log T) = \tilde{O}(\log T), & \alpha = 1, \\ \tilde{\beta} B_0 \frac{\alpha-1}{\alpha-2} = \tilde{O}(1), & \alpha > 1. \end{cases} \tag{5.38}$$

Depending on the convergence rate α , the loss bound $B_\ell(T) := \sup_{\|\xi\| \leq B_0, h \in \mathcal{H}} L(\xi; h)$ scales as $\tilde{O}(1)$ (exponential stability, or $\mathcal{KL}$ -stability with $\alpha > 1$), $\tilde{O}(\log T)$ ($\alpha = 1$), $\tilde{O}(T^{1-\alpha})$ ($\alpha \in (0, 1)$), or, absent any stability assumption at all, $\tilde{O}(L_h^T)$ — exponentially in the horizon T . Faster mixing of the dynamics translates directly into a smaller, more favorable loss bound.

What we have shown is $\sup_{\|\xi\| \leq B_0, h \in \mathcal{H}} L(\xi; h) \leq B_\ell(T)$.

5.6.2 Bounding the Rademacher Complexity

Define $R_n(\ell \circ \mathcal{H}) := \mathbb{E}_{\{\xi_i\} \sim D^n} \mathbb{E}_\sigma \left[\sup_{h \in \mathcal{H}} \frac{1}{n} \sum_i \sigma_i L(\xi_i; h) \right]$, the Rademacher complexity of the trajectory-loss class. By Theorem 5.10 applied at the level of whole trajectories (each ξ_i , not each time step, is one i.i.d. sample), with probability at least $1 - \delta$ over $\{\xi_i\} \sim D^n$, simultaneously for every $h \in \mathcal{H}$,

$$\mathbb{E}_{\xi \sim D} L(\xi; h) \leq \frac{1}{n} \sum_{i=1}^n L(\xi_i; h) + 2R_n(\ell \circ \mathcal{H}) + B_\ell(T) \sqrt{\frac{\log(1/\delta)}{2n}}. \quad (5.39)$$

It remains to bound $R_n(\ell \circ \mathcal{H})$ via section 5.5's covering-number machinery, for a concrete parametric class.

A parametric hypothesis class. Consider $\mathcal{H} = \{h_\theta(x) = h(x; \theta) : \theta \in \Theta \subseteq B_2^k(1) \cdot B_\Theta, \|\theta\|_2 \leq B_\Theta, h \in C^2\}$ — exactly the parametric form $h(\cdot; \theta)$ of Assumption 4.1, now allowed to be an arbitrary twice-differentiable (rather than linear) function of θ .

Step: show $L(\xi; h_\theta)$ is Lipschitz in θ . By the mean-value form of Taylor's theorem, applied coordinatewise to $\theta \mapsto h(x; \theta)$,

$$h(x; \theta_1) - h(x; \theta_2) = \left[\int_0^1 \int_0^1 \frac{\partial^2 h}{\partial x \partial \theta}(\alpha x, \beta \theta_1 + (1 - \beta)\theta_2) d\beta d\alpha \right] \cdot x \cdot (\theta_1 - \theta_2), \quad (5.40)$$

so that, defining $L_{\partial^2 h} := \sup_{\|x\| \leq \gamma(T), \|\theta\| \leq B_\Theta} \|\partial^2 h / \partial x \partial \theta\|_{\mathbb{R}^k \rightarrow \mathcal{L}(\mathbb{R}^k, \mathbb{R}^{n_x})}$ (an operator-norm bound on the mixed second derivative, over the range of states $\|x\| \leq \gamma(T)$ the trajectory actually visits),

$$\|h(x; \theta_1) - h(x; \theta_2)\|_2 \leq L_{\partial^2 h} \cdot \|\theta_1 - \theta_2\|_2 \cdot \|x\|. \quad (5.41)$$

Fixing $x_0 = \xi$ and taking $\ell(h(x_t), x_{t+1}) = \|h(x_t) - x_{t+1}\|_2$ for simplicity (as in (5.36)),

$$\begin{aligned} |L(\xi; h_{\theta_1}) - L(\xi; h_{\theta_2})| &\leq \sum_{t=0}^{T-1} \|h(x_t(\xi); \theta_1) - h(x_t(\xi); \theta_2)\|_2 \\ &\leq L_{\partial^2 h} \|\theta_1 - \theta_2\|_2 \sum_{t=0}^{T-1} \|x_t(\xi)\|_2. \end{aligned}$$

The sum $\sum_t \|x_t\|_2$ is exactly the quantity bounded in (5.37)–(5.38) above (with $\gamma(T)$ set to whichever of $\beta B_0, B_0 L_h^T$, etc. applies) — so, writing $L_x(T)$ for that bound and $L_\theta(T) := L_{\partial^2 h} L_x(T)$,

$$|L(\xi; h_{\theta_1}) - L(\xi; h_{\theta_2})| \leq L_\theta(T) \|\theta_1 - \theta_2\|_2, \quad (5.42)$$

i.e., the trajectory loss is $L_\theta(T)$ -Lipschitz in θ , with the *same* three-way case split on T -scaling as $B_\ell(T)$ above.

Assembling the Dudley bound. Chaining Propositions 5.18 to 5.20 through (5.32):

$$\begin{aligned}
 R_n(\ell \circ \mathcal{H}) &\leq \frac{16}{\sqrt{n}} \int_0^\infty \sqrt{\log N(u; \ell \circ \mathcal{H}, \|\cdot\|_n)} \, du \\
 &\stackrel{(\text{Prop. 5.19})}{\leq} \frac{16}{\sqrt{n}} \int_0^\infty \sqrt{\log N(u/L_\theta(T); \Theta, \|\cdot\|_2)} \, du \\
 &\stackrel{u=L_\theta(T)\varepsilon}{=} \frac{16}{\sqrt{n}} L_\theta(T) \int_0^\infty \sqrt{\log N(\varepsilon; \Theta, \|\cdot\|_2)} \, d\varepsilon \\
 &\stackrel{(\text{Prop. 5.20})}{\leq} \frac{16}{\sqrt{n}} L_\theta(T) \int_0^\infty \sqrt{\log N(\varepsilon/B_\Theta; B_2^k(1), \|\cdot\|_2)} \, d\varepsilon \\
 &\stackrel{\varepsilon=B_\Theta\varepsilon'}{=} \frac{16}{\sqrt{n}} L_\theta(T) B_\Theta \int_0^1 \sqrt{\log N(\varepsilon'; B_2^k(1), \|\cdot\|_2)} \, d\varepsilon' \\
 &\stackrel{(\text{Prop. 5.18})}{\leq} 16 L_\theta(T) B_\Theta \sqrt{\frac{k}{n}} \int_0^1 \sqrt{\log(1 + 2/\varepsilon')} \, d\varepsilon' \\
 &\leq 16 \times 1.3526 \, L_\theta(T) B_\Theta \sqrt{\frac{k}{n}} \\
 &\leq 22 L_\theta(T) B_\Theta \sqrt{\frac{k}{n}}, \tag{5.43}
 \end{aligned}$$

where the substitutions $u = L_\theta(T)\varepsilon$ and $\varepsilon = B_\Theta\varepsilon'$ rescale each integral to match the covering-number property being invoked, and $\int_0^1 \sqrt{\log(1 + 2/\varepsilon')} \, d\varepsilon' \leq 1.3526$ is a numerically evaluated constant.

5.6.3 The Final Generalization Bound

Substituting (5.43) into (5.39):

With probability at least $1 - \delta$ over $\{\xi_i\}_{i=1}^n \sim D^n$, simultaneously for every $h \in \mathcal{H}$,

$$\mathbb{E}_{\xi \sim D} L(\xi; h) \leq \frac{1}{n} \sum_{i=1}^n L(\xi_i; h) + 44 L_\theta(T) B_\Theta \sqrt{\frac{k}{n}} + B_\ell(T) \sqrt{\frac{\log(1/\delta)}{2n}}. \tag{5.44}$$

💡 Key Result 5.6

Both T -dependent quantities in (5.44) — the Lipschitz constant $L_\theta(T)$ and the loss bound $B_\ell(T)$ — obey the *same* three-way split established in Key Result 5.5: $\tilde{O}(1)$ under exponential (or fast polynomial, $\alpha > 1$) stability, $\tilde{O}(T^{1-\alpha})$ under slower polynomial stability, and $\tilde{O}(L_h^T)$ — exponential blow-up in the prediction horizon — with no stability assumption whatsoever. This is the precise, quantitative form of the emphasis flagged at the start of this section: **the stability of the underlying dynamics controls the sample complexity of learning to predict it**, entering the bound not through the dimension k or sample size n (as in the classical, static ERM theory of section 5.4.2) but through the horizon-dependent constants $L_\theta(T)$ and $B_\ell(T)$ multiplying the usual $\sqrt{k/n}$ rate. A system that is easy to *predict from* over a long horizon is, by exactly the same token, easy to *learn to predict from* a correspondingly modest amount of data.

5.7 Further Reading

Chapter 5 is a slim slice of statistical learning theory, chosen to reach the ERM generalization bound of Key Result 5.6 by the shortest reasonably rigorous path; Shalev-Shwartz and Ben-David [SB14] and Wainwright [Wai19] both give comprehensive, textbook-length treatments of everything in this chapter and a great deal more. Hardt and Recht [HR22] cover much of the same material with a different emphasis — prediction as a tool embedded in a larger decision-making pipeline, closer in spirit to the “predicts well” framing that opened this chapter — and is freely available at <https://mlstory.org>. Hoeffding’s inequality (Lemma 5.6) is due to Hoeffding [Hoe63]; McDiarmid’s inequality (Lemma 5.11), and the symmetrization argument built on it, generalizes it substantially and is due to McDiarmid [McD89]. Rademacher complexity as a tool for deriving generalization bounds — the approach this chapter takes throughout sections 5.4 and 5.5 — is due to Bartlett and Mendelson [BM02]; the contraction inequality of Proposition 5.13 is a special case of a substantially more general result in Ledoux and Talagrand [LT91]. Dudley’s entropy integral (Theorem 5.17) and the VC-dimension covering-number bound both appear, with full proofs and considerably more generality, in Wainwright [Wai19, Ch. 5]; VC-dimension itself originates with Vapnik and Chervonenkis [VC68].

Of the other theories of generalization mentioned in section 5.2.3 but not developed here, algorithmic stability is due to Bousquet and Elisseeff [BE02]. Double descent (fig. 5.1) was named and empirically documented across a range of model classes by Belkin et al. [Bel+19]; reconciling it with, rather than merely observing it alongside, the uniform-convergence theory of this chapter is an active area of research, well beyond our scope here.

Finally, the dynamical-systems application of section 5.6 connects directly back to the classical theory of the previous chapter, and to its own Further Reading on the non-asymptotic regime: the horizon-dependent constants $L_\theta(T)$ and $B_\ell(T)$ derived here play exactly the role that the stability of the system played in the AR(1) sidebar of section 4.3.2, now for the ERM/uniform-convergence theory rather than the classical prediction-error estimator.

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